

Unknown Title



[86. Partition List](#)

Medium



Topics

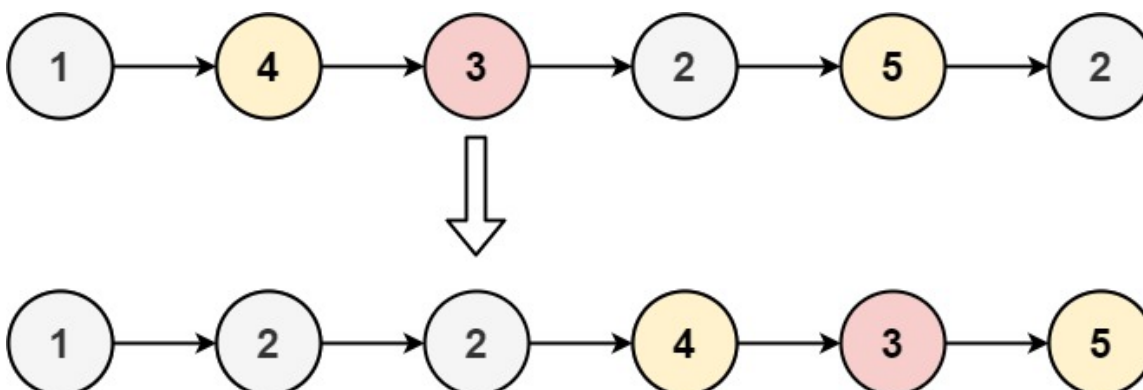


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Given the **head** of a linked list and a value x , partition it such that all nodes **less than** x come before nodes **greater than or equal to** x .

You should **preserve** the original relative order of the nodes in each of the two partitions.

Example 1:



Input: head = [1,4,3,2,5,2], x = 3

Output: [1,2,2,4,3,5]

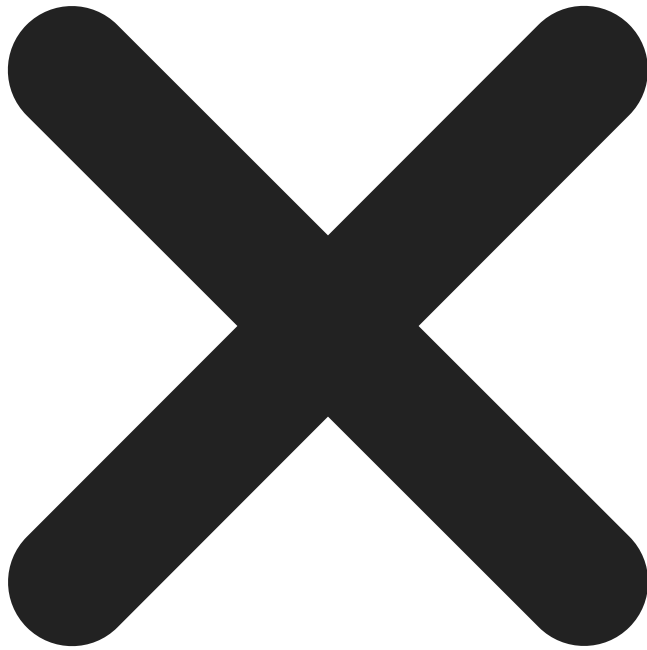
Example 2:

Input: head = [2,1], x = 2

Output: [1,2]

Constraints:

- The number of nodes in the list is in the range [0, 200].
- $-100 \leq \text{Node.val} \leq 100$
- $-200 \leq x \leq 200$



Seen this question in a real interview before?

Yes

No

Accepted

934,071/1.5M

Acceptance Rate

61.5%



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[代发周](#)

Aug 14, 2020

The description for this problem is somewhat very misleading.

I took at least 15 min and tested several test cases so that I understood what it was asking for.

Let me briefly explain it to relieve your pain.

Basically, it asks us:

reorder the sequence by the given number x

The rules are:

1. Any number that is **less than x** has to be before x , and maintain the relative order with those that are less than x but already before x .
e.g. $[3, 4, 1, 2]$, $\text{target} = 4 \rightarrow [3, 1, 2, 4]$, so the order of $[3, 1, 2]$ is maintained.
2. Any number that is **greater than x** but already before x will still be before x , but all of them come after those that are less than x and at the same time maintain their relative order.
e.g. $[3, 6, 5, 4, 1, 2]$ $\text{target} = 4 \rightarrow [3, 1, 2, 6, 5, 4]$
3. Any number that is **greater than x** and after x will only need to maintain their relative order
e.g. $[3, 6, 5, 4, 8, 1, 7, 2]$ $\text{target} = 4 \rightarrow [3, 1, 2, 6, 5, 4, 8, 7]$

Hope this helps those who have a difficult time understanding "**all nodes less than x come before nodes greater than or equal to x .**" like me. 😂😂😂



Fold



1.2K



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[Anuron Das](#)



Aug 14, 2023

It is an interesting problem, at least to me. If I were to explain the problem statement to anyone in the simplest terms, I would go like this:

You are given a linked list and a target number x . You have to reorder the list in such a way that any node with a value **strictly less than x** appears **in the beginning** of the list, such that their relative order **among themselves** is preserved.

For example: $[1, 4, 3, 2, 5, 2]$ and $x=3$. All the numbers **strictly smaller than x** are $[1, 2, 2]$. So preserve only their relative order and add them in the beginning of the list.

After that you have to preserve the relative order of all the **remaining elements** as well, and after that you can return the list **head**.

For example, let us consider the above example only, in $[1, 4, 3, 2, 5, 2]$ and $x=3$, after removing all the numbers **strictly less than x** , the remaining list is $[4, 3, 5]$. So their relative order is preserved accordingly.

In the end just concatenate the two and return the result.

So if the problem statement is clear, then one can appreciate the importance of **two pointers** approach, as this problem involves the use of **two pointers** on linked lists. I hope the reader finds this helpful.



Fold

Tip



119



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[Daniil](#)

Jan 10, 2023

tip: Don't forget to clear the Next of the last element of the list if you are spreading elements across two lists. I tried for half an hour to figure out why my solution fails with an error until I realized that I was returning a list with a loop



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Tip



87



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[Homer](#)

Feb 07, 2015

"For example,

Given 1->4->3->2->5->2 and x = 3,

return 1->2->2->4->3->5."

In the return, why does 4 come before 3 if it's larger???

Also, my code fails this test:

"Input: {2,1}, 1

Output: {1,2}

Expected: {2,1}"

If x is 1, why would the expected output have 2 come before 1?

Thanks.

^

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[Abhishek shende](#)



Jul 05, 2021

[328. Odd Even Linked List](#)

above problem is quite similiar to this problem after trying this you can solve that too.



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[An-Wen Deng](#)

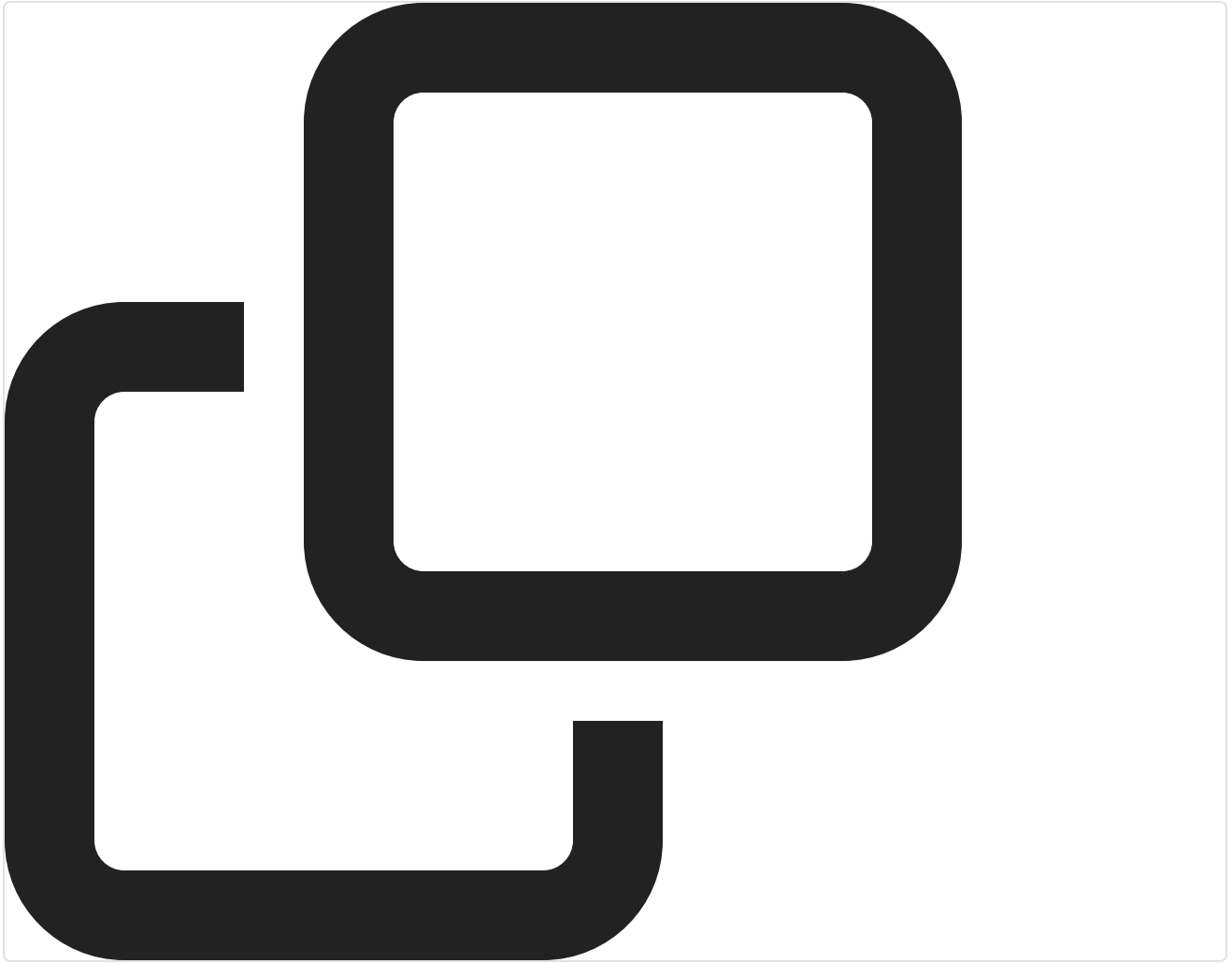


Aug 14, 2023

Divide the given list head into 2 linked lists, greater & smaller. Then connect these 2 lists and return the result. One more hint:

The whole process shown by modifying submitted code in playground

```
1<3:smaller
smaller:0->1->4->3->2->5->2
greater:0
-----
4>=3: greater
smaller:0->1->4->3->2->5->2
greater:0->4->3->2->5->2
-----
3>=3: greater
smaller:0->1->4->3->2->5->2
greater:0->4->3->2->5->2
-----
2<3:smaller
smaller:0->1->2->5->2
greater:0->4->3->2->5->2
-----
5>=3: greater
smaller:0->1->2->5->2
greater:0->4->3->5->2
-----
2<3:smaller
smaller:0->1->2->2
greater:0->4->3->5->2
-----
ans:
1->2->2->4->3->5
```



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38



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[Abdelrhman](#)

Feb 15, 2019

Input: head = 1->4->3->2->5->2, x = 3

Output: 1->2->2->4->3->5

why 4 is exist before than 3 not after ?



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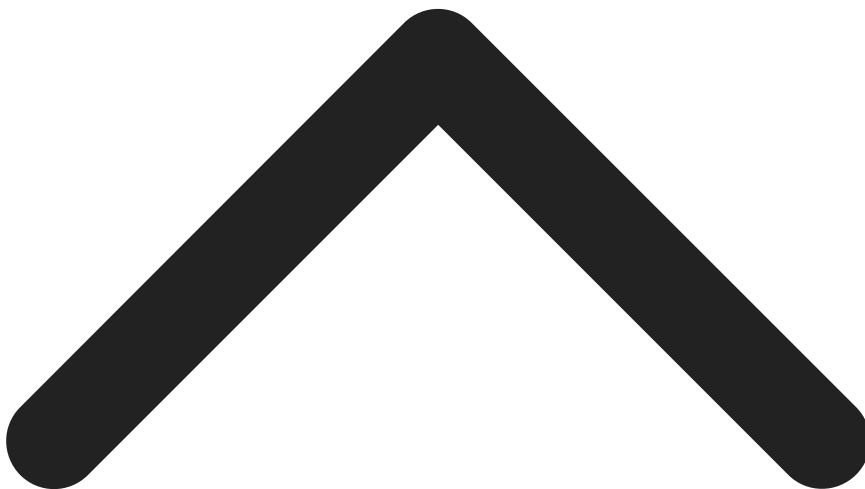


[Prisha](#)



Aug 15, 2023

If handling the swapping with a double pointer feels tricky, here's a hint: consider using two linked lists. In one list, gather elements less than x, and in the other, store elements greater than x. Then, simply connect these two lists together. Happy coding!



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16



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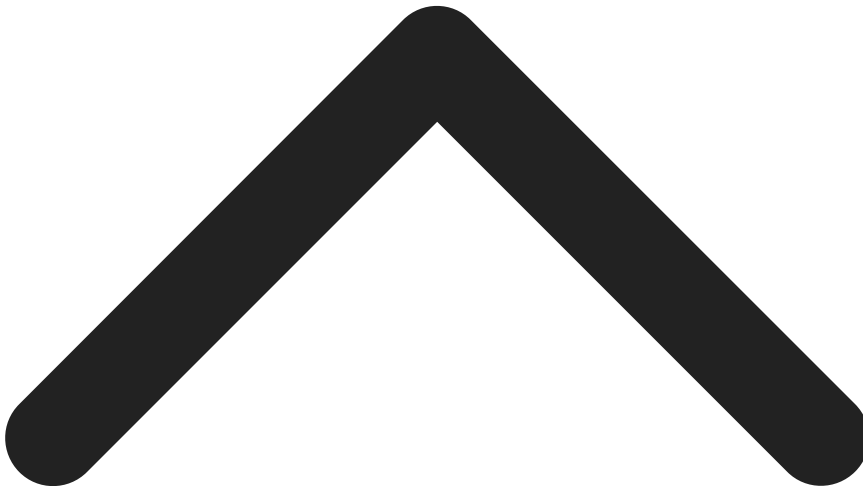
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[valige7091](#)

Feb 15, 2021

All the problem asks one to do is loop through a list and add each element to one of two lists, before combining the two then returning.



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[Sujal Goswami](#)



Aug 15, 2023

no leetcode no partition today



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